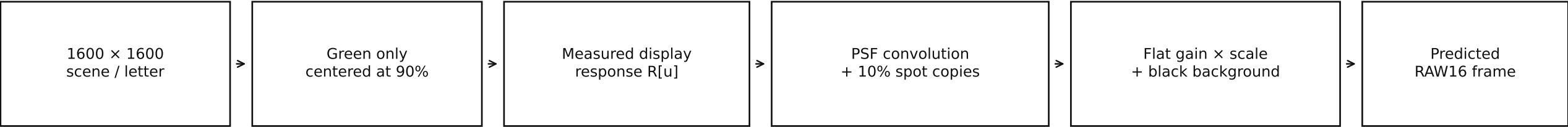


A compact model from display pixels to RAW16

A linear-intensity approximation for the green display channel and a fixed lenslet camera setup



$$\hat{y} = d + s g \odot \{0.90 C_h(R[u]) + 0.10 L(R[u])\}$$

d : monitor-black map · g : white-field pixel gain · h : measured 2-D PSF · L : weighted lenslet copies

DISPLAY	CAMERA	CALIBRATION	ASSUMPTION
1920 × 1080 Green channel; 90% fullscreen	ASI174MM · RAW16 80 ms · gain 0	120 spot centers Measured response, background, flat field	Incoherent light Intensity contributions add linearly

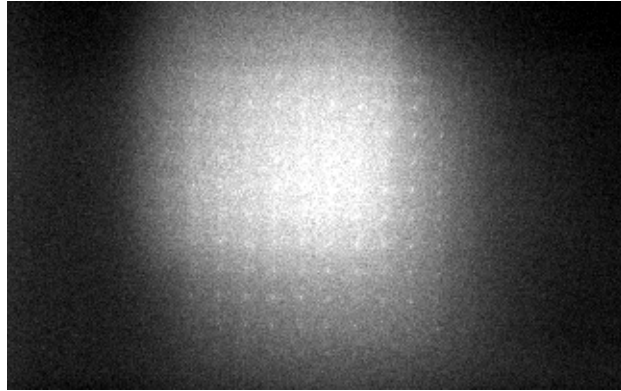
Model terms are shared across the alphabet and 20 square scenes; scene references are used only in the evaluation figures.

Fitted constants: signal scale 0.777 · PSF sampling 1.30× / 1.00× · spot mix 0.10

What is fitted in the forward model?

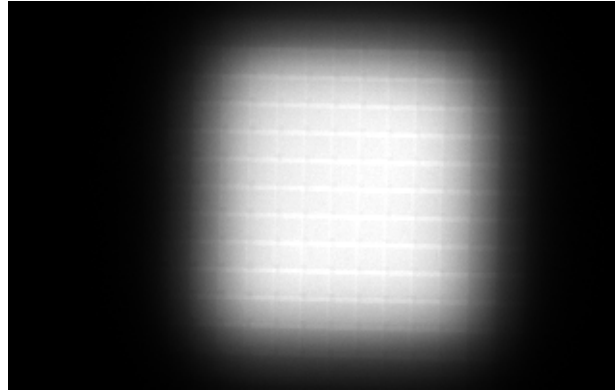
Each map/curve comes from calibration captures; detector maps are shown with robust contrast limits.

1 Monitor-black background



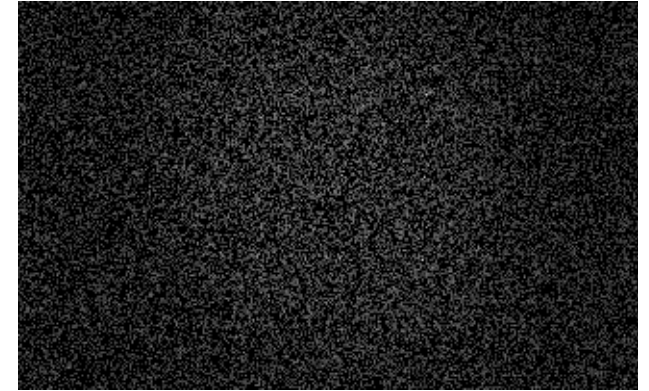
RAW16 DN; includes ambient light

2 Flat-field pixel gain



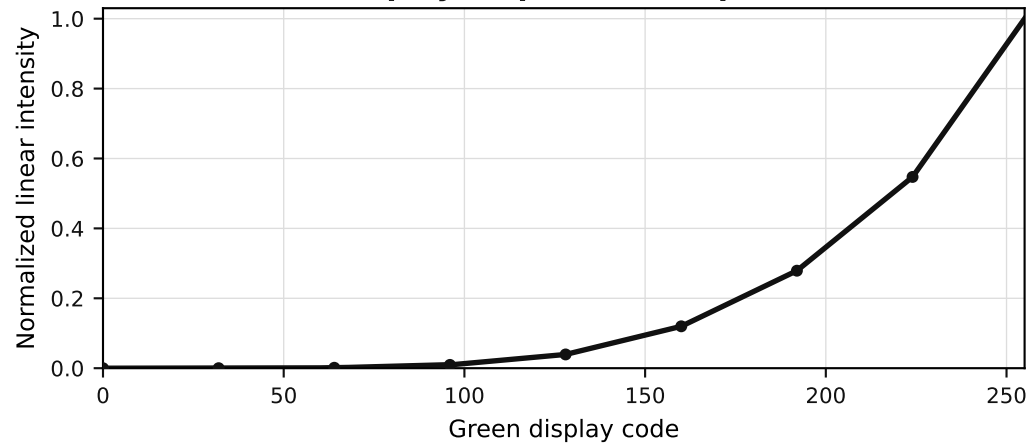
White field – background

3 Measured 2-D PSF



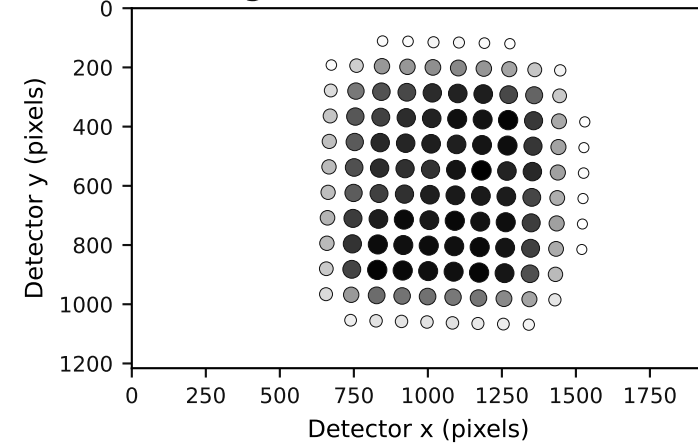
Square-root display reveals weak spots

4 Display response lookup table



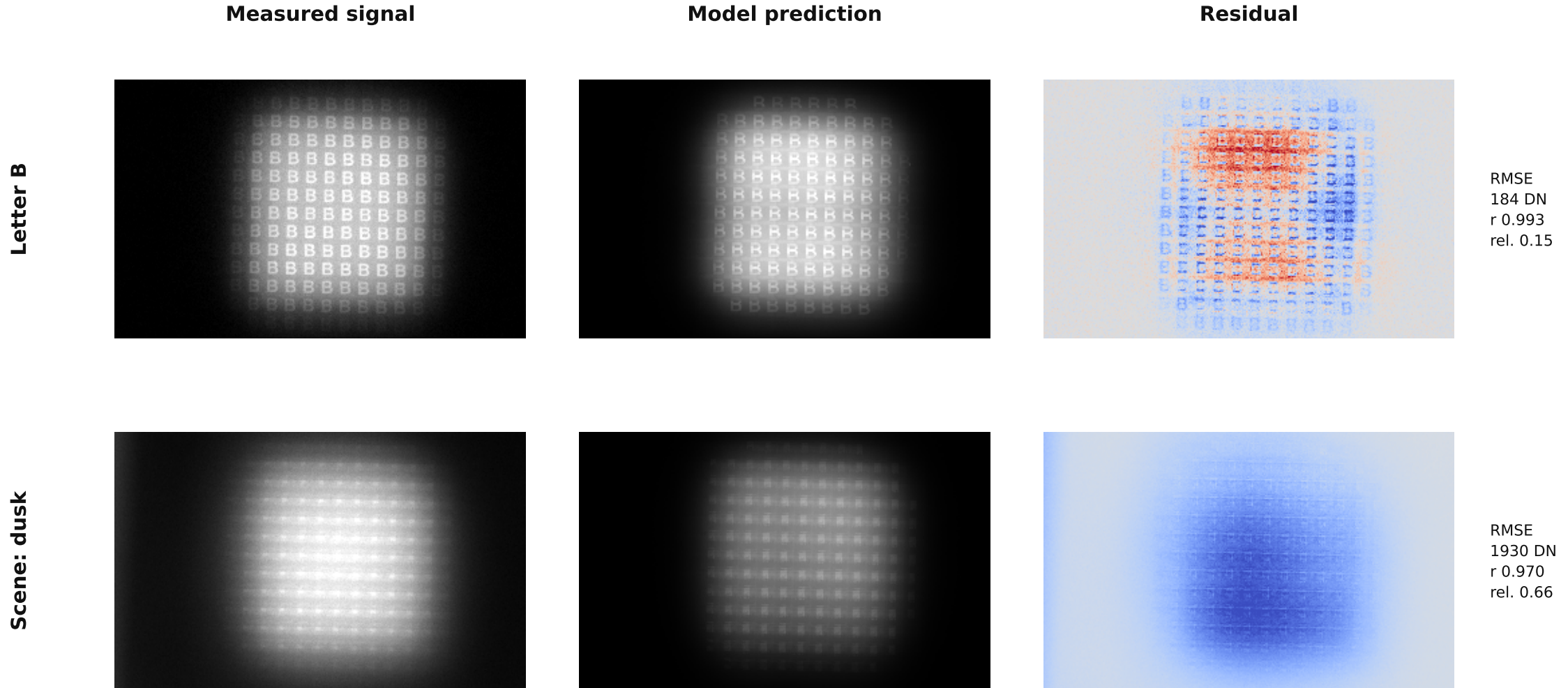
Spot scale $|m_x|=0.117$, $|m_y|=0.128$ · PSF sampling $x=1.30$, $y=1.00$ · scalar signal scale=0.777

5 Weighted lenslet centers (120)



Forward prediction against measured RAW16

The two rows show an asymmetric letter and a textured scene; residual = prediction – measurement.



Signal panels subtract the fitted monitor-black map; both use a shared scale within each row.

Residual panels use a symmetric scale; RAW16 measurements are at 80 ms exposure and gain 0.

Measurement-only reconstruction: letters

A, B, F, and O reconstructed by flat-field correction, spot-cell folding, and median averaging.

Letter A



Displayed



Reconstructed

Letter B



Displayed



Reconstructed

Letter F



Displayed



Reconstructed

Letter O



Displayed



Reconstructed

Each reconstruction uses only its RAW16 measurement (120 copies); the displayed image is included only for visual comparison.

Measurement-only reconstruction: scenes (1/2)

Green-only display at 90%; reference and reconstruction are shown as centered square crops.

Dusk

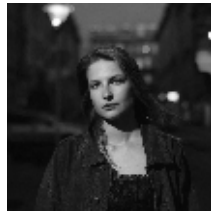


Displayed

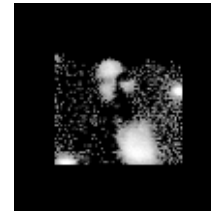


Reconstructed

Citystreet



Displayed

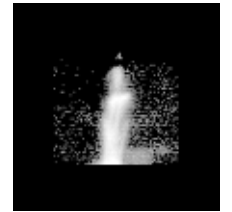


Reconstructed

Falls



Displayed

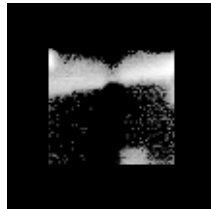


Reconstructed

Forest



Displayed



Reconstructed

Harbor



Displayed

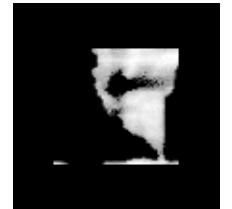


Reconstructed

London



Displayed



Reconstructed

Spot-fold reconstructions use RAW16 measurements only; references are for comparison. Dusk: C. Stone · Citystreet: Artem Mihailov · Falls: Chris Briggs · Forest: Fabian Kleiser · Harbor: Lukas Rychvalsky · London: Dust Studio

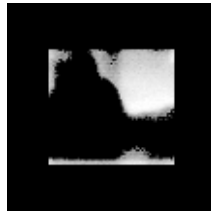
Measurement-only reconstruction: scenes (2/2)

Green-only display at 90%; reference and reconstruction are shown as centered square crops.

Mountain



Displayed

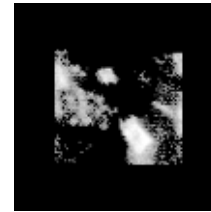


Reconstructed

Produce



Displayed

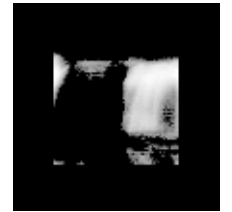


Reconstructed

Rocks



Displayed

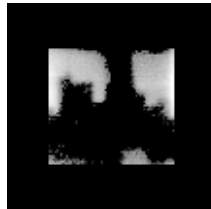


Reconstructed

Skyline



Displayed

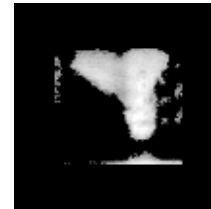


Reconstructed

Street



Displayed



Reconstructed

Vineyard



Displayed



Reconstructed

Spot-fold reconstructions use RAW16 measurements only; references are for comparison. Mountain: Aleks Dahlberg · Produce: Maria Wang · Rocks: Gabe Hobbs · Skyline: Rowan Freeman · Street: Alice · Vineyard: Trent Erwin